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Architectural Evolution of nopCommerce

Scenario C - Omnichannel Commerce Core

TP1 Group 02

Presented by:
- António Alberto 114622
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- João Varela 113780

SCENARIO CHOICE

The Problem

VerdeMart uses nopCommerce as its web storefront only

Order state needs to reflect stores and warehouses

Operations cannot stop when external systems lag

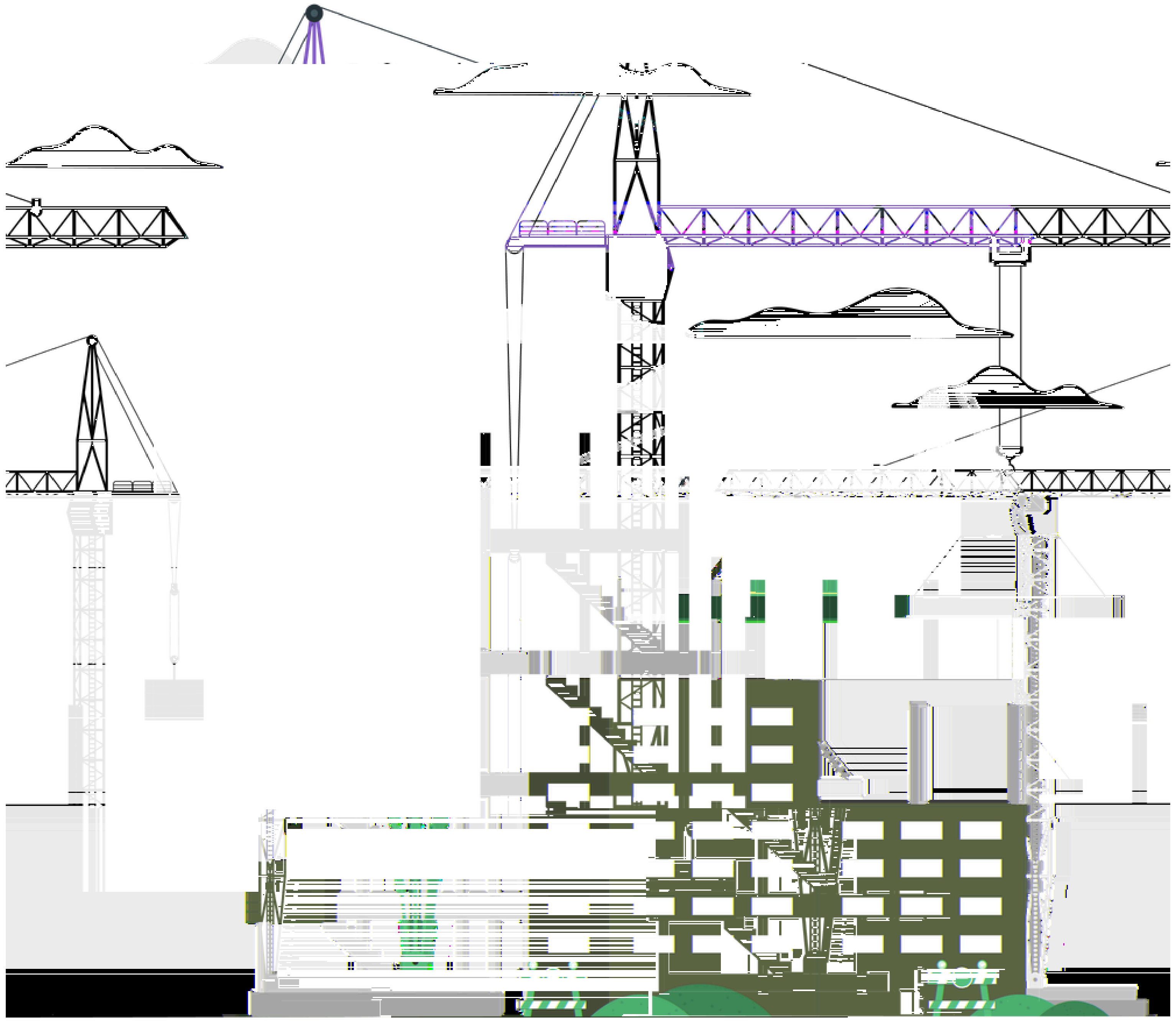
The Choice

Map directly onto existing nopCommerce domain

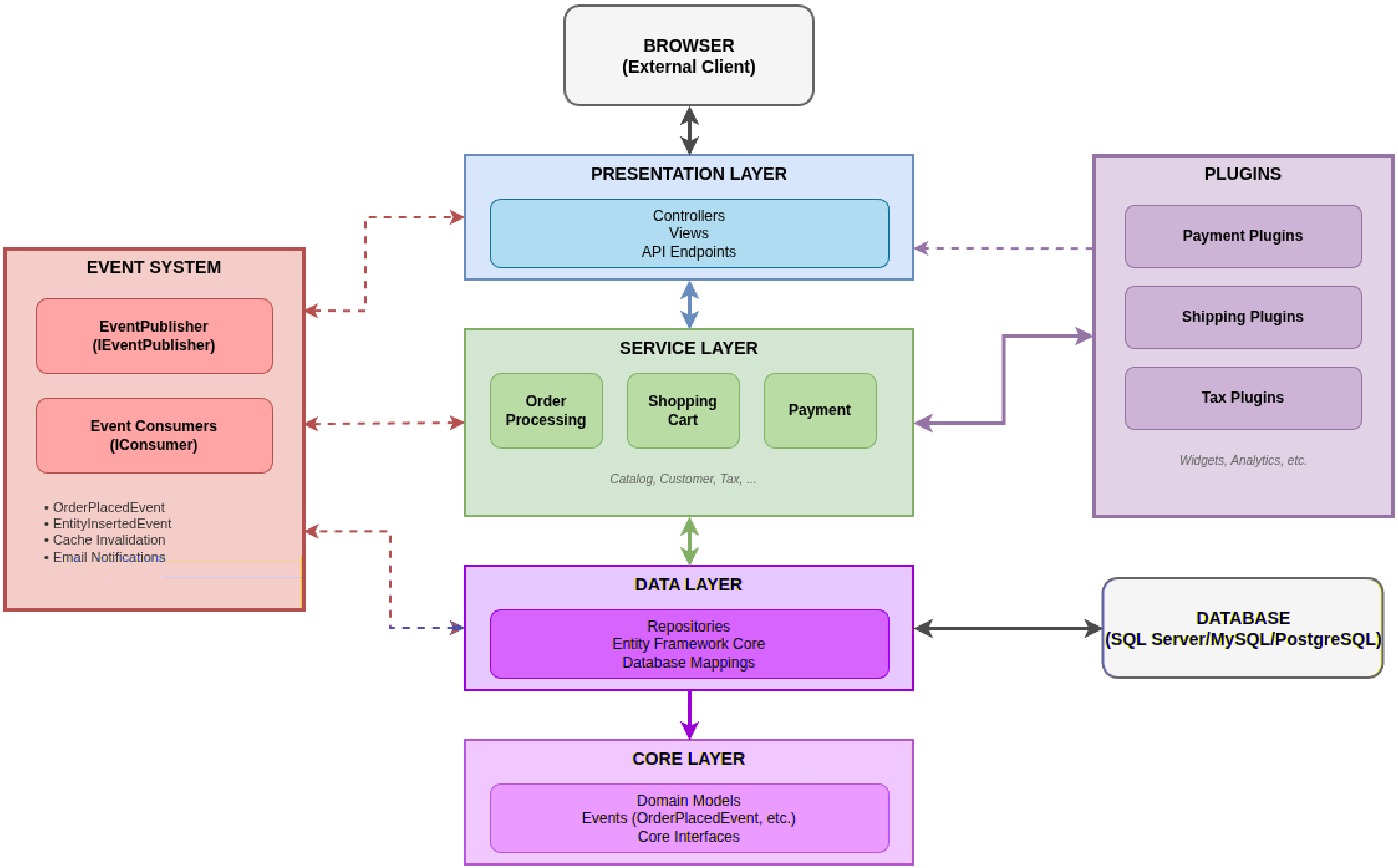
Turn nopCommerce into a Commerce Core

Evolve, don’t Invent

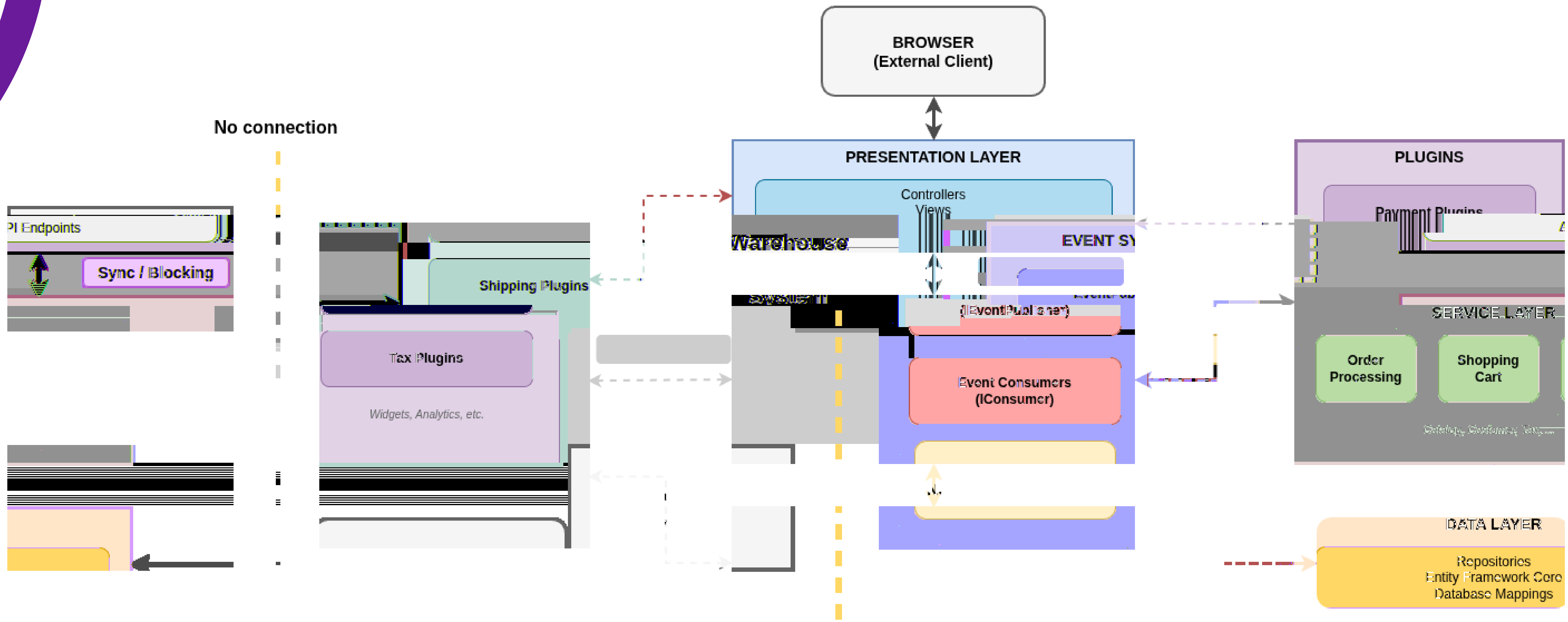
CURRENT STATE ANALYSIS



nopCommerce Communication Architecture



nopCommerce Communication Architecture



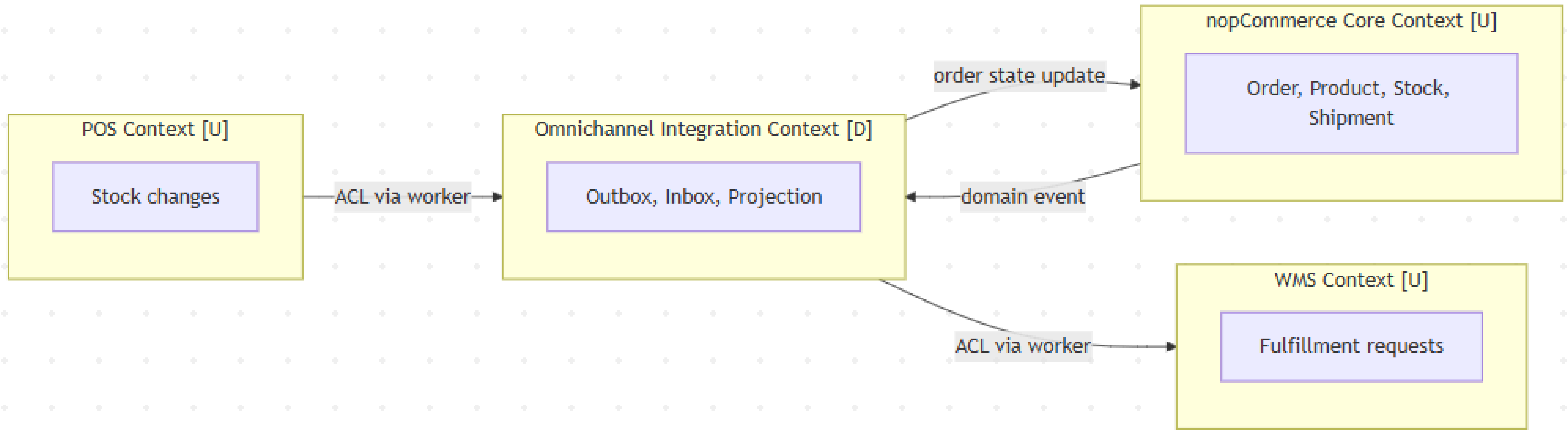
DOMAIN AND BOUNDARY MODEL

6 Subdomains

- **Commerce Core:** checko , orders, c s omers, ca alog, pricing, pa men s a s.
- **Inventory Visibility:** prod c s ock projec ion, areho se-specific q an i ies, s ale-s a e de ec ion.
- **Fulfillment Coordination:** bridge from placed order o areho se processing.
- **Integration Reliability:** o bo , inbo , re ries, DLQ, idempo enc .
- **Warehouse Operations:** e ernal WMS beha ior (accep /rejec /dela /con radic).
- **Store Operations / POS:** s ock mo emen s ha origina e o side he eb channel.

DOMAIN AND BOUNDARY MODEL

4 Bounded contexts



QUALITY ATTRIBUTE SCENARIOS

- **QA-1 Resilience & Recovery:** WMS 503 for 30s → backlog drains ≤ 60s after recovery & 0 orders pending > 5 min.
- **QA-2 Consistency:** duplicate POS event → rejected ≤ 50ms & 0 duplicate fulfillments.
- **QA-3 Traceability:** support investigates pending order → outbox → MQ → worker → projection & resolution ≤ 3 admin clicks.
- **QA-4 Performance:** OrderPlacedEvent fired → outbox row written ≤ 100ms & 0 sync HTTP calls in checkout.
- **QA-5 Cross-Channel Visibility:** WMS marks order shipped → nopCommerce status updated ≤ 10s & 0 state divergences.

CHOSEN FRAMEWORK

ADD

Attribute-Driven Design

- Iterative, driver-first development
- Refines the system iteratively based on drivers

ACDM

Architecture-Centric Design Method

- Go/no-go Decisions
- Supports evaluations and experiments

ADM

Architecture Development Method

- Enterprise oriented method for planning
- Migration planning, governance and transforming sequencing

CHOSEN FRAMEWORK

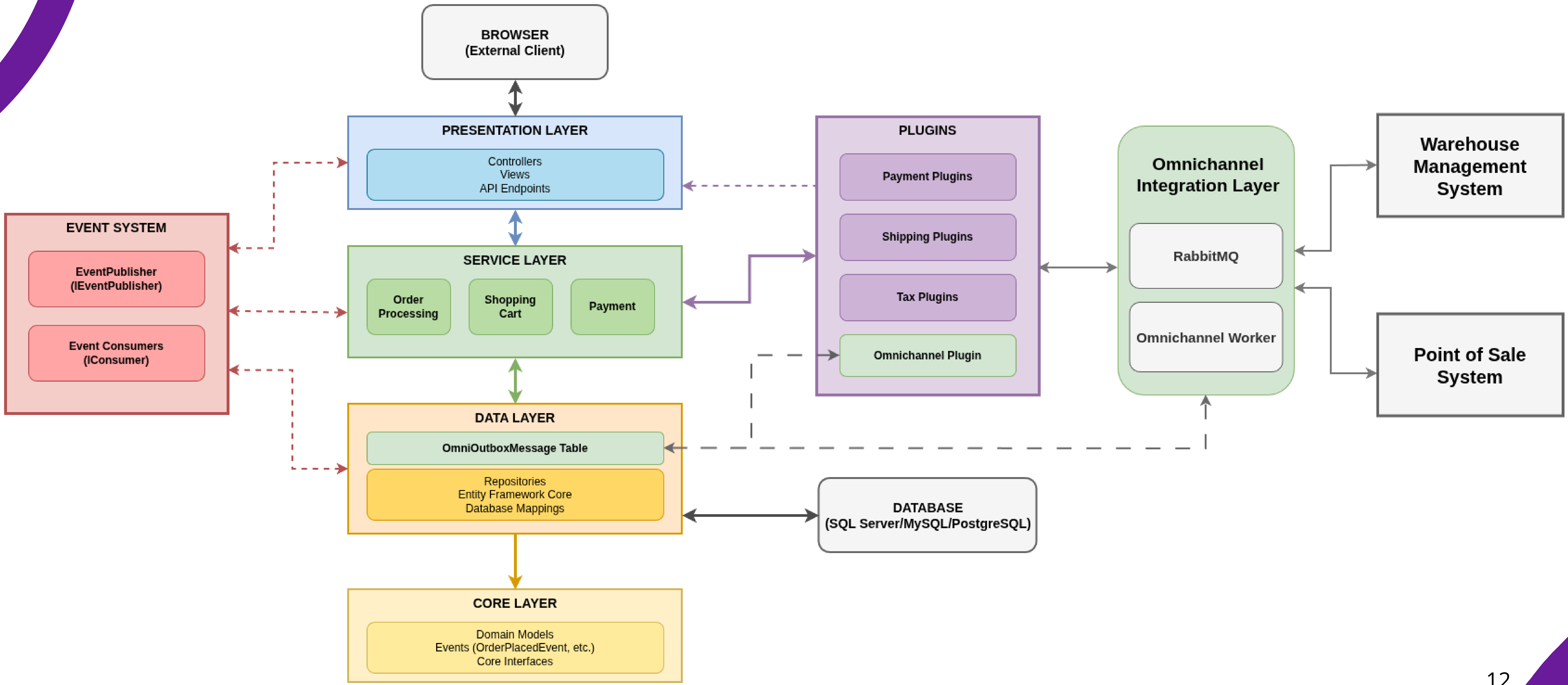
Framework	FIT	Reason

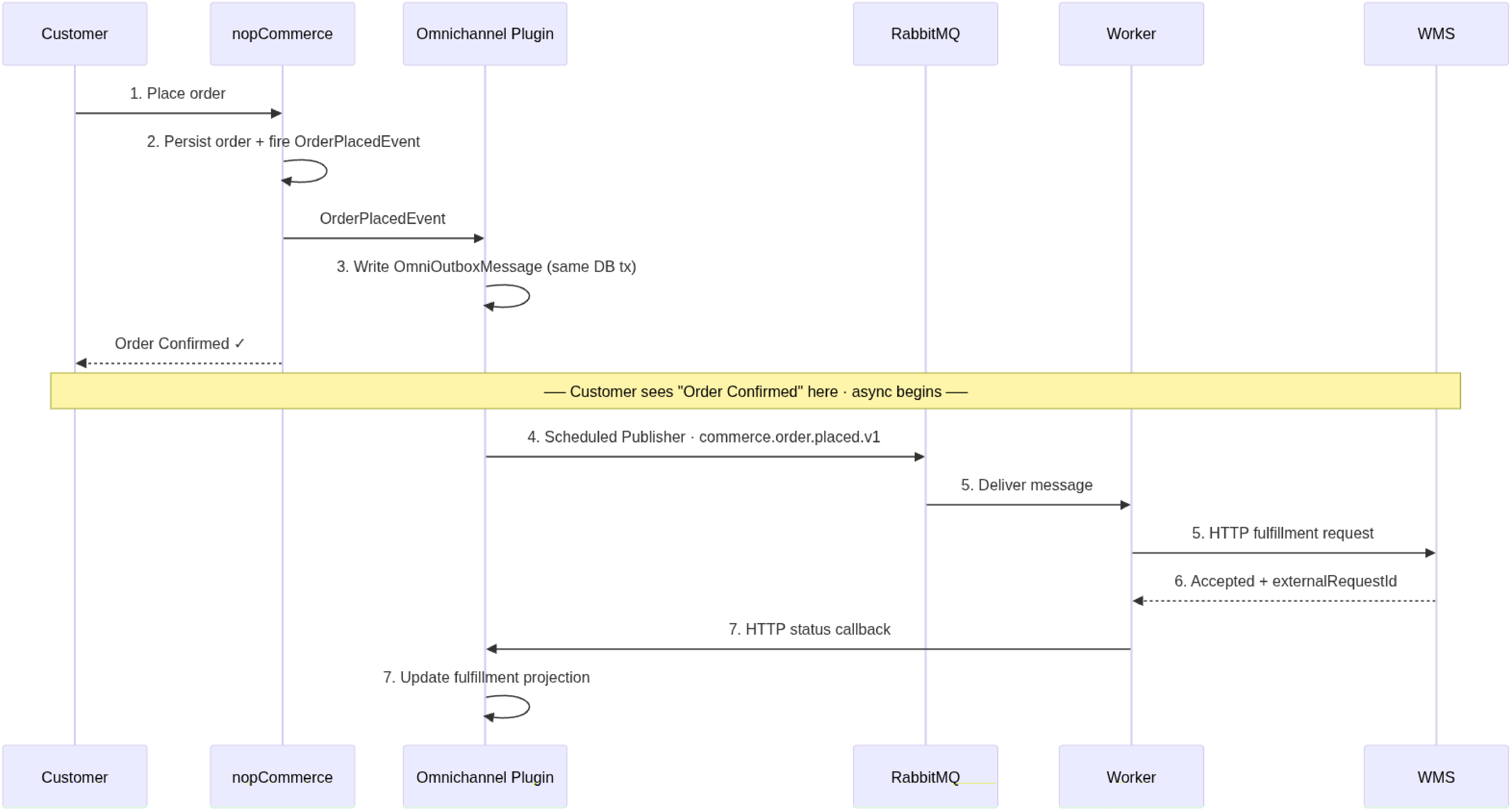
What we use from each

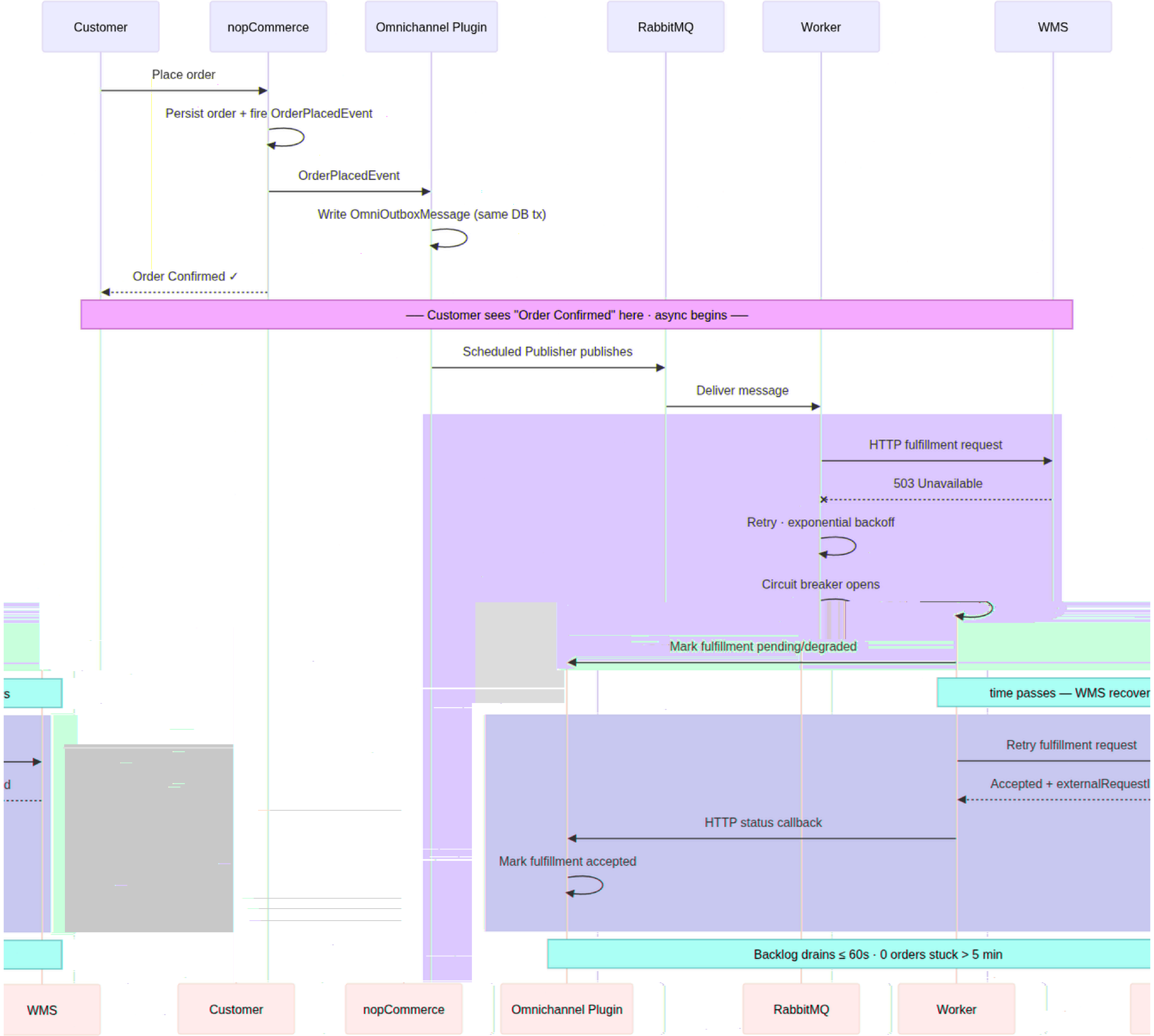
Design Interactions + Tactics
Go / no-go checks
Phase-F Migration Roadmap table

TARGET ARCHITECTURE

nopCommerce Communication Architecture




Customer
nopCommerce
Omnichannel Plugin
RabbitMQ
Worker
WMS
1. Place order
2. Persist order + fire OrderPlacedEvent
OrderPlacedEvent
3. Write OmniOutboxMessage (same DB tx)
Order Confirmed ✓
── Customer sees "Order Confirmed" here · async begins ──
4. Scheduled Publisher · commerce.order.placed.v1
5. Deliver message
5. HTTP fulfillment request
6. Accepted + externalRequestId
7. HTTP status callback
7. Update fulfillment projection
Customer
nopCommerce
Omnichannel Plugin
RabbitMQ
Worker
WMS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Place order
Persist order + fire OrderPlacedEvent
OrderPlacedEvent
Write OmniOutboxMessage (same DB tx)
Order Confirmed ✓
── Customer sees "Order Confirmed" here · async begins ──
Scheduled Publisher publishes
Deliver message
HTTP fulfillment request
503 Unavailable
Retry · exponential backoff
Circuit breaker opens
Mark fulfillment pending/degraded
time passes — WMS recover
Retry fulfillment request
Accepted + externalRequestI
HTTP status callback
Mark fulfillment accepted
Backlog drains ≤ 60s · 0 orders stuck > 5 min
WMS
Customer
nopCommerce
Omnichannel Plugin
RabbitMQ
Worker

ARCHITECTURAL DECISIONS

ADR	Decision	Alternative Considered	WHY?
0003	Outbox + RabbitMQ	Sync checkout-to-WMS call	Checkout stays independent from WMS failures
0005	No shared DB boundary	Worker SQL Access	Boundaries stay explicit and deployable
0008	3-ID correlation	Timestamp-only tracing	Delayed orders stay explainable end-to-end
0009	Reject microservices extraction	Extract services by bounded context	Coordination is the architectural problem, decomposition just adds overhead
0010	Reject distributed tracing infra	OpenTelemetry / Jaeger stack	ADR-8's 3-ID correlation satisfies QA-3 with less infrastructure

RISKS & VALIDATION PLAN

Key risks

- Checkout accidentally depends on WMS availability
- Duplicate messages create duplicate fulfillment
- Stock projection becomes misleading
- Scope grows beyond a selective evolution
- Plugin integration is harder than expected

EVOLUTION ROADMAP

FEASIBILITY SPIKE

Feasibility Spike

"Can a placed order start an asynchronous omnichannel workflow without making checkout depend on WMS/POS availability?

Plugin can write durable outbox row and return.

Checkout stays independent from WMS/POS availability.

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